

# Amar Dhillon

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## EDUCATION

**University of Maryland, College Park**  
*B.S. in Computer Science, Minor in Statistics; GPA: 3.5*

College Park, MD  
*Expected May 2028*

## CERTIFICATIONS

**Amazon Web Services (AWS) Certified Developer - Associate**  
*Amazon Web Services*

Nov 2024 – Nov 2027  
*Digital Badge*

## EXPERIENCE

**Software Development Engineer Intern**  
*Amazon Web Services*

Sep 2026 – Dec 2026  
*Redmond, WA*

- Writing backtested Java capacity-planning logic that sets upper and lower bounds on the EC2 capacity Amazon Redshift reserves for patching to balance availability against cost

**Software Engineering Intern**  
*Capital One*

Jun 2026 – Aug 2026  
*Richmond, VA*

- Split a Claude skill into sub-skills backed by Python search and validation tools to stop open-ended model search, which cut LLM workflow generation time **65%** to under 15s with no quality loss in stress tests
- Precomputed popular workflows from a 270K+ row PostgreSQL table into a Redis-cached category tree to keep lookups within the database's permission and rate limits, which became the Claude skill's main search tool
- Rebuilt a legacy test-data workflow tool as a React and TypeScript Electron app with a visual React Flow editor and Jest-tested validation to make editing smoother, which **90%** of 30+ surveyed developers preferred

**Software Engineer (Contract)**  
*UMD App Development Contracting (Client: Amazon Leo)*

Jan 2026 – Present  
*College Park, MD*

- Built an EventBridge-scheduled LLM-as-judge check for a production Amazon Bedrock classifier that rates research documents by importance to trigger fine-tuning only when its grades shift past a threshold
- Reduced end-to-end latency **34%** for a pipeline routing 6,000+ scraped research documents to inference by replacing S3 staging hops with Kinesis streams feeding Lambda
- Co-built a research tool that scores and aggregates topics to show Amazon Leo's public-policy staff what's popular, which runs in dev on AWS Fargate with GitHub Actions CI/CD and AWS CDK infrastructure-as-code ready for production

**Machine Learning Research Intern**  
*UMD Institute for Advanced Computer Studies (UMIACS)*

Dec 2025 – Present  
*College Park, MD*

- Built a pipeline that overlays AnyGrasp's candidate grasps on camera images in Open3D to let a vision-language model pick robot grasps for a PhD-led ICLR project, and tuned its prompts to favor grasps that avoid hitting the table

**Co-Founder, Software Engineer**  
*TerpLabs*

Jun 2025 – Present  
*College Park, MD*

- Building RAG meal suggestions for the official UMD app with SentenceBERT vector search in ChromaDB to feed GPT-4o mini real scraped recipes, which internal testers rated satisfactory for **93%** of meals
- Built Redis caching in front of the Prisma ORM queries for Tortuga, the studio's class scheduler with 20,000+ visits in its first 30 days, to reduce database load and latency

## PROJECTS

**IEX Order Book & Matching Engine** | *C++17, STL, POSIX, Linux perf*

Jul 2026 – Aug 2026

- Built a C++17 order book and matching engine to replay **50.9M** real IEX exchange messages without ever resyncing, which agreed with the exchange on 99,122 of 99,123 fills and 100% of top-of-book prices

## TECHNICAL SKILLS

**AI/ML:** LLMs (Claude, Amazon Bedrock, GPT-4o mini), LLM agents and tool use (Claude skills), retrieval-augmented generation (RAG), embeddings and vector search (SentenceBERT, ChromaDB), LLM-as-a-judge evaluation, model fine-tuning, vision-language models, prompt engineering, Open3D

**Languages:** Python, Java, C++, C, TypeScript, JavaScript, SQL

**Cloud & Data:** AWS (Lambda, Kinesis, EventBridge, Fargate), AWS CDK, monitoring (CloudWatch, X-Ray), REST APIs, PostgreSQL, Redis, OpenSearch

**Engineering:** full-stack apps (React, Electron), data structures and algorithms, object-oriented design, Docker, GitHub Actions (CI/CD), Linux, unit testing (Pytest, Jest), Git, uv